Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P2-Model)** **Date: 29-03-2026**

Time: 02.30Pm to 05.30Pm **GTA-21** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**AC**	2	**ACD**	3	**ABD**	4	**ABC**	5	**AD**	6	**BD**
7	**AD**	8	**AD**	9	**0.22**	10	**20**	11	**1**	12	**2**
13	**2**	14	**4.8**	15	**B**	16	**D**	17	**B**	18	**C**

CHEMISTRY

19	**AB**	20	**AB**	21	**BCD**	22	**ABD**	23	**CD**	24	**ABCD**
25	**ABC**	26	**AB**	27	**2**	28	**1**	29	**142**	30	**744**
31	**5**	32	**3**	33	**B**	34	**C**	35	**D**	36	**B**

MATHEMATICS

37	**BC**	38	**ABD**	39	**BC**	40	**BD**	41	**AC**	42	**AC**
43	**AD**	44	**AB**	45	**4**	46	**8**	47	**0.6**	48	**8**
49	**105**	50	**61**	51	**C**	52	**B**	53	**D**	54	**B**

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SOLUTIONS

PHYSICS

1. Angular velocity of earth $\omega_1 = \frac{\pi}{12} rad/hr$

Angular velocity of satellite $\omega_2 = \frac{\pi}{3} rad/hr$

$(\omega_2 - \omega_1)t = 2\pi$ t = 8 hours

$\rightarrow$ for option 'B'

$T^2 \alpha r^3$ T' = 48 hours

$\omega' = \frac{\pi}{24} hours$ $(\omega' = \omega_1)t = 2\pi$ t = 48 hours

2. $m = \frac{f}{f-u}$ ____(i) $\frac{h_i}{h_0} = \frac{f}{f-u}$

$h_i = h_0\left(\frac{f}{f-u}\right)$ ____(ii)

$v = u\frac{f}{u-f}$ ____(iii)

Differentiate equation (ii) and (iii) with respect time then we will get velocity of image.

3. $\vec{r} = \alpha t^3 \hat{i} + \beta t^2 \hat{j}$ $\vec{v} = \frac{d\vec{r}}{dt} = 3\alpha t^2 \hat{i} + \beta t \hat{j}$

$\vec{a} = \frac{d^2\vec{r}}{dt^2} = 6\alpha t\hat{i} + 2\beta\hat{j}$

At t = 1

A) $\vec{v} = 3 \times \frac{10}{3} \times 1\hat{i} + 2 \times 5 \times 1\hat{j} = 10\hat{i} + 10\hat{j}$

B) $\vec{L} = \vec{r} \times \vec{p}$ $\left(\frac{10}{3} \times 1\hat{i} + 5 \times 1\hat{j}\right) \times 0.1\left(10\hat{i} + 10\hat{j}\right) = -\frac{5}{3}\hat{k}$

C) $\vec{F} = m \times \left(6 \times \frac{10}{3} \times 1\hat{i} + 2 \times 5\hat{j}\right) = \left(2\hat{i} + \hat{j}\right)N$

D) $\vec{\tau} = \vec{r} \times \vec{F} = \left(\frac{10}{3}\hat{i} + 5\hat{j}\right) = \left(2\hat{i} + \hat{j}\right) = +\frac{10}{3}\hat{k} + 10\left(-\hat{k}\right) = -\frac{20}{3}\hat{k}$

4. As we know that $v = v_0 e^{-4t}\cos(\omega_d t + \theta_0)$

Here $y = \frac{R}{2L}; \omega_d^2 = \omega_0^2 - \left(\frac{R}{2L}\right)^2$ $\omega_0^2 = \frac{1}{LC}$

For oscillatory motion ω_d should be real for under damping

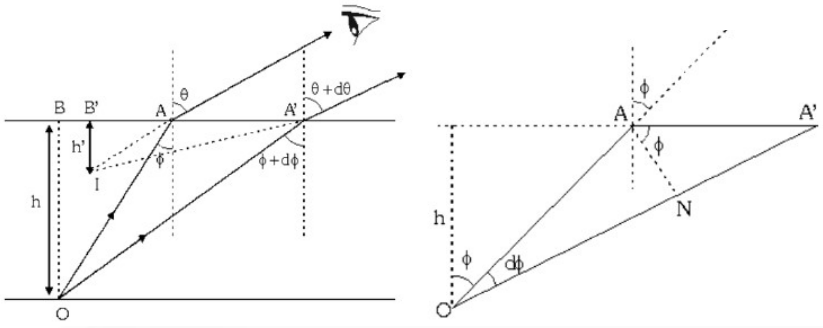
$\therefore \frac{1}{LC} - \frac{R^2}{4L^2} > 0 \Rightarrow R < 2\sqrt{\frac{L}{C}}$ $\Rightarrow R < 2\sqrt{\frac{4 \times 10^{-3} \times 10}{16 \times 10^{-6}}}$

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$R<100$

$\left.\begin{array}{l} R=100 \rightarrow \text{Critical damping} \\ R>100 \rightarrow \text{overdamping} \end{array}\right\}$ in both cases there are no oscillatory

5.



$$AO=\frac{h}{\cos\phi} \qquad AN=(OA)(d\phi)$$

$$AN=(OA)(d\phi)=\frac{h}{\cos\phi}.d\phi$$

$$AA'=\frac{AN}{\cos\phi}=\left(\frac{h}{\cos\phi}\right)d\phi.....(1)$$

Similarly from triangle IAA' $\qquad AA'=\left(\frac{h'}{\cos^2\phi}\right)d\phi.....(2)$

Equating eqn (1) and (2) $\qquad \left(\frac{h}{\cos^2\phi}\right)d\phi=\left(\frac{h'}{\cos^2\phi}\right)d\phi$

$$h'=h\left[\frac{\cos^2\phi}{\cos^2\phi}\right]\left(\frac{d\phi}{d\phi}\right)$$

$$\sin\theta=\mu\sin\phi.....(4)$$

$$\cos d\theta=\mu\cos\phi\, d\phi$$

$$\frac{d\phi}{d\phi}=\frac{\cos\theta}{\mu\cos\phi}.....(5)$$

From eqn (5) & (1)

$$h'=\frac{h}{\mu}\left[\frac{\cos^3\theta}{\cos^3\phi}\right].....(6)$$

$$\cos\phi=\sqrt{1-\sin^2\phi}=\sqrt{1-\frac{\sin^2\phi}{\mu^2}} \qquad \cos^3\phi=\left[1-\frac{\sin^2\phi}{\mu^2}\right]^{3/2}$$

$$\cos^3\phi=\frac{\left[\mu^2-\sin^2\phi\right]^{3/2}}{\mu^2}.....(7)$$

From eqn (6) & (7) $\qquad h'=\frac{\mu^2\cos^3\phi}{\left[\mu^2-\sin^2\phi\right]^{3/2}}$

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6.



$mv = mv_1 + Mv_2 \;\&\; v_2 - v_1 = v \qquad \therefore\; v_2 = \dfrac{2v}{\left(\dfrac{M}{m}+1\right)}$



$Mv_2 = Mv_3 + mv_4 \;\&\; v_4 - v_3 = v_2 \qquad \therefore\; v_4 = \dfrac{4Mv}{(M+m)^2}$

7. Case – I : (Smooth barrel) $\qquad h_1 = \dfrac{1}{2}gt^2 \;\&\; E_1 = \dfrac{1}{2}mv_1^2$

Case – II : (Rifled barrel)

$h_2 = \dfrac{1}{2}gt^2 \;\&\; E_2 = \dfrac{1}{2}mv_1^2 + \dfrac{1}{2}I\omega^2 \left(I = kmR^2\right)$

Let horizontal distance of the target be D. $\qquad \therefore\; D = v_1\sqrt{\dfrac{2h_1}{g}} = v_2\sqrt{\dfrac{2h_2}{g}}$

Also, $\omega = \dfrac{2\pi vN}{L}$ (L = length of barrel)

Also since energy released in firing is same, hence

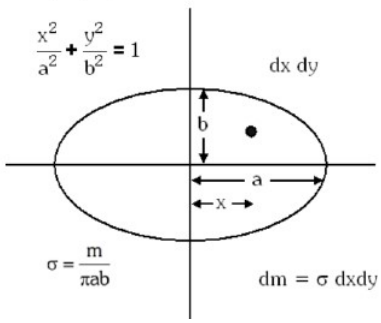
$E_1 = E_2 = E \qquad \Rightarrow \dfrac{1}{2}mv_1^2 = \dfrac{1}{2}mv_2^2 + \dfrac{1}{2}m\omega^2 \;\Rightarrow\; v_1^2 = v_2^2 + \dfrac{4\pi^2N^2R^2v_2^2}{L^2}$

Also, $v_1^2 = \dfrac{D^2g}{2h_1} \;\&\; v_2^2 = \dfrac{D^2g}{2h_2} \qquad \therefore \dfrac{1}{h_1} - \dfrac{1}{h_2} = \dfrac{4\pi^2N^2kR^2}{L^2h_2}$

$\therefore \dfrac{h_2 - h_1}{h_1h_2} = \dfrac{4\pi^2N^2kR^2}{L^2h_2} \qquad \therefore\; L = 2\pi N\sqrt{k}R\sqrt{\dfrac{h_1}{h_2 - h_1}}$

$E_{rot} = \dfrac{1}{2}mv_1^2 - \dfrac{1}{2}mv_2^2 \qquad \therefore \dfrac{E_{rot}}{E_t} = \dfrac{v_1^2 - v_2^2}{v_1^2} = 1 - \dfrac{h_1}{h_2}$

8.



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About $x \Rightarrow dIx = dmy^2$

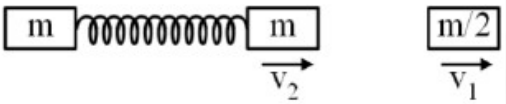
$$dIx = \frac{4m}{\pi ab}\int_0^a dx\int_0^y dy\, y^2 \qquad = \frac{4m}{\pi ab}\int_0^a dx\frac{y^3}{3}$$

Where $y^2 = b^2\left(1-\frac{x^2}{a^2}\right)$ $\qquad = \frac{4m}{\pi ab}\int_0^a b^3\left\{1-\frac{x^2}{a^2}\right\}^{3/2} dx$

$$= \frac{4m}{\pi ab}\frac{b^3}{a^3}\int_0^a \left\{a^2-x^2\right\}^{3/2} dx \qquad = \frac{4m}{\pi}\frac{b^2}{a^4}\int_0^{\pi/2} a^3\cos^3\theta \times a\cos d\theta$$

$$= a\sin\theta; dx = a\cos\theta d\theta = \frac{4mb^2}{\pi}\int_0^{\pi/2}\cos^2 d\theta \qquad = \frac{4mb^2}{\pi}\left[\frac{1}{8}-\frac{1}{16}\right]\pi = \frac{mb^2}{4}$$

9. Just after collision, the system will lock like,



$$mv_0 = \frac{m}{2}v_1 + mv_2 \;\&\; v_1 - v_2 = v_0$$

$$\therefore\ 2v_0 = \frac{3mv_1}{2} \Rightarrow v_1 = \frac{4v_0}{3} \Rightarrow v_2 = \frac{v_0}{3}$$

$$\therefore \frac{1}{2}kx_{\max}^2 = \frac{1}{2}\frac{m}{2}\left(\frac{2v_0}{3}\right)^2 \left[\frac{1}{2}kx_{\max}^2 = \frac{1}{2}\mu v_{rel}^2\right] \Rightarrow x_{\max} = \sqrt{\frac{4mv_0^2}{18k}} = \sqrt{\frac{2mv_0^2}{9k}}$$

10. Velocity of inner belt $= \omega R$

$\therefore$ Time after which it will be at same mark $= \frac{2L+2\pi R}{v} = \frac{8\pi R}{\omega R} = \frac{8\pi}{\omega}$

Velocity of middle belt $= \omega^2 R$

$\therefore$ Time after which it will be at same mark $= \frac{2L+2\pi 2R}{v} = \frac{10\pi}{\omega}\frac{R}{2R} = \frac{5\pi}{\omega}$

Velocity of outer belt $= \omega^3 R$

$\therefore$ Time after which it will be at same mark $= \frac{2L+2\pi 2R}{\omega 3R} = \frac{12\pi}{3\omega} = \frac{4\pi}{\omega}$

$\therefore$ They will be at same mark after $20\left(\frac{2\pi}{\omega}\right)$ time.

11. $Q = \frac{P^2}{2M}+\frac{P^2}{2M} = \frac{P^2}{2m}\left(\frac{m}{M}+1\right)$ $\qquad Q = E_\alpha\left(\frac{m}{M}+1\right)$ $\qquad \therefore\ E_\alpha = \frac{QM}{M+m}$

12. $E = \frac{n^2h^2}{8mL^2}$ $\qquad \Delta E = \frac{(4-1)h^2}{8mL^2} = \frac{3h^2}{8mL^2}$

$\phi = \frac{h^2}{8mL^2}$ $\qquad (KE)_{\max} = \Delta E - \phi = 2\frac{h^2}{8mL^2}$

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$= 2 \times$ Energy of electron at n = 1 $\Rightarrow x = 2$

13. $$\frac{\int \vec{B}.\vec{dl}\,CDA}{\int \vec{B}.\vec{dl}\,ABC} = \frac{\frac{\mu_0 I}{2} + \frac{\mu_0 I}{2\pi}\frac{\pi}{6}\times 2}{\frac{\mu_0 I}{2} - \frac{\mu_0 I}{2\pi}\left(\frac{\pi}{6}\right)\times 2} = 2$$

14. $$E = \frac{\rho}{3\varepsilon_0}\frac{R}{2} - \frac{\rho(R/2)^3}{3\varepsilon_0 R^2} = \frac{\rho R}{8\varepsilon_0}$$

15. $$f_0 = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}},\ f_1 = \frac{1}{4L_0}\sqrt{\frac{T_0}{\mu}} = \frac{f_0}{2}\ f_2 = \frac{1}{4L_0}\sqrt{\frac{T_0}{2\mu}} = \frac{f_0}{2\sqrt{2}}$$

$$f_3 = \frac{1}{4L_0}\sqrt{\frac{T_0}{3\mu}} = \frac{f_0}{2\sqrt{3}}\ f_4 = \frac{1}{2L_0}\sqrt{\frac{T_0}{4\mu}} = \frac{f_0}{4}$$

16. $$f_0 = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}}$$

For string – 2 $$\frac{3}{2\times\frac{3L_0}{2}}\sqrt{\frac{T_0}{\mu}} = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}} \qquad T_2 = \frac{T_0}{2}$$

For string – 3 $$\frac{5}{2\times\frac{5L_0}{4}}\sqrt{\frac{T_3}{3\mu}} = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}} \qquad T_3 = \frac{3T_0}{16}$$

For string – 4 $$\frac{14}{2\times\frac{7L_0}{4}}\sqrt{\frac{T_4}{4\mu}} = \frac{1}{2L_0}\sqrt{\frac{T_0}{\mu}} \qquad T_4 = \frac{T_0}{16}$$

17. $$W_{1\to2\to3} = P_0V_0 = RT_0$$

$$\Delta U_{1\to2\to3} = nC_V(T_f - T_i) = 1\times\frac{3}{2}(3P_0V_0 - P_0V_0) = 3RT_0$$

$$\Delta Q_{1\to2\to3} = \Delta U + W = 4RT_0$$

$$\Delta Q_{1\to2} = nC_P\Delta T = \frac{5}{2}(2P_0V_0 - P_0V_0) = \frac{5}{2}RT_0$$

18. $$W_{1\to2\to3} = W_{1\to2} + W_{2\to3} = R\frac{T_0}{3}\ln\frac{2V_0}{V_0} + 0 = \frac{RT_0}{3}\ln 2$$

$$\Delta U_{1\to2\to3} = \Delta U_{1\to2} + \Delta U_{2\to3} = 0 + nC_V\left(T_0 - \frac{T_0}{3}\right) = 0 + 1\times\frac{3R}{2}\times\frac{2T_0}{3} = RT_0$$

$$\Delta Q_{1\to2} = \int T\,dx = \int T\left(\frac{3R}{2T}dt + \frac{R}{2}dV\right) = \frac{3R}{2}\int dT + \int_{v_0}^{2v_0}\frac{RT}{V}dV = 0 + RT\ln 2$$

$$\Delta Q_{2\to3} = \int T\,dx = \frac{3R}{2}\int_{T_0/3}^{T_0} dT + \int\frac{RT}{V}dV = \frac{3R}{2}\times\frac{2T_0}{3} + 0 + RT_0$$

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CHEMISTRY

19. NCERT P. NO 451

20. NGP possible from the Br. A and B gives meso product but C and D gies racemic ring

21.

	$A^- +$	H^+	$\rightleftharpoons HA$
Initial point at $2T$	**50**	**10**	**-**
Equivalence Point	**40**	**-**	**10**

$K_a = \dfrac{[H^+][A^-]}{[HA]} = \dfrac{10^{-9}\times 40}{10} = 4\times 10^{-9}$ $n_{HCl} = 50 \Rightarrow V_{HCl} = 100\,\text{ml}$ $\therefore [HA] = \dfrac{50}{200} = 0.25M$

$pH = \dfrac{1}{2}(pKa - \log C) = \dfrac{1}{2}\left(9 - \log^4 - \log 0.25\right)$ $n_{H^+\text{excess}} = 1\times 0.5 = 0.5\,\text{mm}$ $[H^+] = \dfrac{0.5}{201} = 2.5\times 10^{-3}M$

22. Shortest distance between two T.V. is $\dfrac{a}{2}$.

23. Calcium forms nitrolim

24. Conceptual

25. $A = (CH_3)_2 SiCl_2; B = (CH_3)_3 SiCl; D = CH_3SiCl_3$

26. $\left[Ca^{3+}\right]\left[\vec{C}\equiv\vec{C}\right]$ compared to $H_2C = C = CH_2$ has shorter bond length $O_2^+ < O_2 < O_2^- BL$ (based on m.o.t)

27. $X = 0, Y = 2$

28. Only d. It is $Ar\,SN^1$ mech.

29. $H_2(g) + \dfrac{1}{2}O_2(g) \rightarrow H_2O(l); \Delta H = -286\,kJ$

$-286 = \left(B.E._{H-H} + \dfrac{1}{2}\times 498\right) - 2\times B.E._{O-H} - 44$ ____(1)

$H_2(g) + O_2(g) \rightarrow H_2O_2(l);\ \Delta H = -188\,kJ$

$-188 = (B.E._{H-H} + 498) - (2\times B.E._{O-H} + B.E._{O-O}) - 53$ ____(2)

From (1) – (2), we get $B.E._{O-O} = 142\,kJ/mol$

30. $\Delta T_f = K_f.m = 1.86\times \dfrac{\left(\dfrac{3.6}{180} + \dfrac{3.6}{60}\right)}{200}\times 1000 = 0.744°C$ ∴ F.P of solution = - 0.744°C

31. x = 4, y = 4, z = 3, w = 4

32. As_2S_3, Sb_2S_3, SnS_2 = Soluble in YAS , CuS, CdS and PbS = Insoluble in YAS

33, 34. I – Q; II – RP; III – V; IV – TS

35,36. **I)** $A = A_0 - kt$ $A^2 = A_0^2 - 2kA_0t + k^2t^2$

$M = -2kA_0 + 2k^2t \Rightarrow M = -2kA_0$ at $t = 0$ & $M = 0$ $t \geq \dfrac{A_0}{k}$

IV) $\dfrac{1}{A} = \dfrac{1}{A_0} + kt$ $\dfrac{1}{A^2} = \dfrac{1}{A_0^2} + \dfrac{2kt}{A_0} + k^2t^2$ $M = \dfrac{2k}{A_0} + 2k^2t$

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MATHEMATICS

37. For $\alpha = 1$, $\lim_{n\to\infty} x_n = 1$.

For $\alpha \in (0,1)$ $\lim_{n\to\infty} x_n = 0$

For $\alpha \in \left(1, \frac{3}{2}\right)$ $\lim_{n\to\infty} x_n = \infty$

38. Subtracting pair wise we get,

$(x-1)^2 = 0, (x-1)(2x+1) = 0, (x-1)(x+2) = 0$

So third equation becomes $(x-1)(x+2)\left(x+\frac{1}{2}\right) = 0$

Hence $\alpha = -\frac{3}{2}, \beta = -\frac{7}{2}, \gamma = -2$

39. $G = \left(X_1^{f_1} X_2^{f_2} \ldots\ldots X_n^{f_n}\right)^{\frac{1}{N}}$ where $N = \sum f_i$

$$\log G = \frac{1}{N}\sum f_i \log(X_i) = \frac{1}{N}\sum f_i \log(x_i + M)$$

$$= \frac{1}{N}\sum f_i \left(\log M + \log\left(1 + \frac{x_i}{M}\right)\right)$$

$$= \frac{1}{N}\sum f_i \left(\log M - \frac{x_i}{M} - \frac{1}{2}\left(\frac{x_i}{M}\right)^2 + \frac{1}{3}\left(\frac{x_i}{M}\right)^3 \ldots\right)$$

$$= \log M + \frac{\sum f_i x_i}{NM} - \frac{\sum f_i x_i}{2NM^2} = \log M - \frac{b^2}{2M^2}$$

$$G = M.e^{-\frac{b^2}{2M^2}} = M\left(1 - \frac{b^2}{2M^2}\right)$$

Squaring, $G = M\left(1 - \frac{b^2}{M^2}\right) = M^2 - b^2$.

40. Consider $xy = c^2$

$A(t_1), B(t_2), C(t_3), D(t_4)$

Let circle be $x^2 + y^2 + 2gx + 2fy = 0$

Putting 't', $ct^4 + 2gt^3 + 2ft + c = 0$

$\sum t_1 t_2 = 0$ So $t_1t_2 + t_2t_4 + t_1t_4 + \frac{t_1 + t_2 + t_4}{t_1t_2t_4} = 0$

Tangents at t_1 & t_2 intersects at $\left(\frac{2ct_1t_2}{t_1 + t_2}, \frac{2c}{t_1 + t_2}\right)$

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So locus of the point is $\Rightarrow x^2+y^2+2ct_4x+\frac{2c}{t_4}y=0$

Centre $@\left(-ct_4,-\frac{c}{t_4}\right)$ passing through $(0,0)$

41. Conceptual
42. JEE ADV 2025 P1
43. JEE ADV 2025 P1
44. JEE ADV 2025 P2
45. $f(x)=x\ell nx$
46. As director circle passes through $(\pm5,\pm3)$, so centre of the conic must be at origin. So if one focus is at (x,y) then the other focus must be at $(-x,-y)$.

    Hence $(x-5)(-x-5)=(y-3)(-y-3)=(\text{semi minor axis})^2$

    So eqn of locus is $x^2-y^2=16$

    Length of $LR=8$.
47. Orthocentre $=\left(-a,a\left(t_1+t_2+t_3+t_1t_2t_3\right)\right)$

    $=(-a,k)$ $\quad$ $=(-5\cos\theta,4\sin\theta)$

    Locus is $\frac{x^2}{25}+\frac{y^2}{16}=1$ $\quad$ $e=\sqrt{1-\frac{16}{25}}=\frac{3}{5}=0.6$
48. Describing the problem for standard parabola. Let $L(t_1),L'(t_2),M(t_3),M'(t_4)$

    Constant terms of circle eqns with LL' and MM' as diameter are $a^2\left(t_1^2t_2^2+4t_1t_2\right)$ and $\left(t_3^2t_4^2+4t_3t_4\right)$. Since both chords intersect on axis, $t_1t_2=t_3t_4$ Hence radical axis passes through $(0,0)$ i.e, vertex of parabola $\alpha=1,\beta=2$.
49. JEE ADV 2025 P1
50. Let circle be $(x-t)^2+y^2=r^2,t>0,r\geq t+1$

    So $t^2\leq(r-1)^2$

    $(r-1)^2+11^2\geq r^2\Rightarrow r\leq61$
51. JEE ADV 2023 P1
52. JEE ADV 2024 P1

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